

Curriculum Vitae

PERSONAL INFORMATION

Rajib Baran Roy

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WORK EXPERIENCE

November, 2016 - Till

Assistant Professor

Department of Electrical and Electronic Engineering, Primeasia University, Star Tower 12 Kemal Ataturk Avenue, Banani, Dhaka - 1213, Bangladesh

- Conducted classes in the undergraduate level, supervise undergraduate projects of the students
- Conduct courses EEE 1111: Electrical Circuits I, EEE 1211: Electrical Circuit II, EEE 2217: Energy Conversion I, EEE 2319: Energy Conversion II, EEE 2311: Power System I, EEE 4121: Power System II, EEE 4125: Power Electronics, EEE 4221: Power System Protection, EEE 4223: High Voltage Engineering, EEE 4225: Power System Operation and Control, EEE 4368: Measurement and Instrumentation, EEE 4329: Renewable Energy Technology
- Was the convenor of curriculum development and accreditation committee
- Was a member of IQAC (Institutional Quality Assurance Cell)
- Was a member of research initiative committee for conducting seminars on research skills
- Was a convenor of IEEE student's section for collaborating with IEEE, Bangladesh

March, 2009 - October, 2016

Senior Lecturer

Department of Electrical and Electronic Engineering, University of Information Technology and Sciences, Jamalpur Twin Tower, Baridhara View, GA-37/1, Progoti Sarani, Baridhara J-Block, Dhaka – 1212, Bangladesh

- Conducted classes in the undergraduate level, mentoring undergraduate projects of the students
- Conducted courses EEE 2111: Electronics I, EEE 2213: Electronics II: Energy Conversion I, EEE 2319: Energy Conversion II, EEE 2311: Power System I, EEE 4121: Power System II, EEE 4125: Power Electronics, EEE 4127: Power Transmission and Distribution, EEE 4368: Measurement and Instrumentation, EEE 4329: Renewable Energy Technology
- Was a Member of IEB (Institute of Engineer's, Bangladesh) Accreditation committee
- Was the coordinator IEEE student's section for collaborating with IEEE, Bangladesh

July, 2004 - January, 2006

Lecturer

Department of Electrical and Electronic Engineering, University of Information Technology and Sciences, Baridhara View, GA-37/1, Progoti Sarani, Baridhara J-Block, Dhaka – 1212, Bangladesh

- Instructed undergraduate courses and oversee student projects at the undergraduate level
- Instructed courses, EEE 1111: Electrical Circuits I, EEE 1211: Electrical Circuit II, EEE 2217: Energy Conversion I, EEE 2319: Energy Conversion II, EEE 2322: Electrical Wiring Design, EEE 4127: Power Transmission and Distribution, EEE 4368: Measurement and Instrumentation, EEE 4329: Renewable Energy Technology
- Was a member of curriculum development and accreditation committee

EDUCATION

2022-Till

Doctor of Philosophy in Electrical Engineering (continuing)

Thesis Title: Grid Impacts of Fast Charging Electric Ferry

Central Queensland University, School of Engineering and Technology, Australia

2006-2008

Master of Science in Sustainable Energy System and Management

Thesis Title: 'Evaluation of socio-economic impacts of rural electrification through dissemination of photovoltaic technology as compared to conventional electrification strategies in Bangladesh'.

University of Flensburg, Germany

Core Courses: Solar Energy, Wind Energy, Regional Energy Planning, Trading of Energy, LEAP, Biomass, Biogas, Mini-grids for Rural Electrification, Research Design and Data Collection

SKILL, AWARD, TRAINING

Computer skills

- Python, C++, MATLAB and Simulink
- Electronics Workbench, PSpice, HOMER (basic), DIgSILENT PowerFactory
- OpenDSS, PSCAD, Power World Simulator, RETScreen, LEAP, Proteus, Microsoft Office

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Scholarships and Awards	▪ Recipient of RTP (Research Training Program) scholarship for doing PhD in Australia ▪ Recipient of DAAD (German Academic Exchange Program) scholarship for doing Master's in Germany ▪ Recipient of University scholarship in undergraduate program ▪ Recipient of merit scholarship in college and school
Training	▪ Completed basic training on Homer (Pro) and Homer (Grid) software ▪ Got training on microcontroller-based design and implementation ▪ Completed training on application of PLC for industrial application ▪ Obtained basic training on operation and maintenance of power station and substation ▪ Completed training on Introduction to Renewable Hydrogen, Pure Energy Centre, Shetland Islands, Scotland, UK ▪ Participated in the field study on renewable energy in the Shetland Islands, Scotland, UK which is jointly organized by Highlands and Islands Energy Company (HICEC), UK and the University of Flensburg, Germany

RESEARCH PUBLICATIONS

Publications Impact factor Google Scholar: Citations-492, h-index-13, i10 index-16
 Scopus: Citatons-356, h-index-7

Book	[1] M. R. Amin and R. B. Roy , Electrical Power System Analysis: Control of Voltage and Reactive Power, ISBN: 978-3-659-39552-9, LAP LAMBERT Academic Publishing, Germany
Journal Publications	[1] R.B.Roy , S. Alahakoon and P.J.V. Rensburg, 'Impact of Coordinated Electric Ferry Charging on Distribution Network Using Metaheuristic Optimization', Energies, vol.18, issue 11, pp. 4846, Vol. 9, 2025. [2] R.B.Roy , S. Alahakoon, P.J.V. Rensburg and S.J. Arachchillag, 'Impact Analysis on Distribution Network due to Coordinated Electric Ferry Charging', IET Energy Systems Integration, Willy, 2024. [3] R.B.Roy , S. Alahakoon, P.J.V. Rensburg and S.J. Arachchillag, 'Impact Analysis of Uncoordinated Electric Ferry Charging on Distribution Network', e-prime-Advances in Electrical Engineering, Electronics and Energy, Elsevier, 2024. [4] S. Alakoon, R.B.Roy and S.J. Arachcillage, 'Optimizing Load Frequency Control in Standalone Marine Microgrids using Meta-Heuristic Techniques', Energies, vol.16, issue 13, pp. 4846, Vol. 9, 2023. [5] R.B.Roy , S. Alahakoon, S.J. Arachchillag and S. Rahman, 'Optimizing Dynamic Ferry Loads with Intelligent Power Management', Energy Reports vol. 9, pp. 5952-5963, December, 2023. [6] R.B.Roy , M. Rokonnuzaman, N. Amin, M.K.Mishu, S. Alakoon, S. Rahman, M.Nadaraja, K.S. Rahman and M. Shakeri, 'A Comparative Performance Analysis of ANN Algorithms for MPPT Energy Harvesting in Solar PV System', IEEE Access, Vol. 9, pp. 102137 – 102152, July, 2021. [7] F.A. Hossain, M. Rokonuzzaman, N. Amin, J. Zhang, M.K. Mishu, W.S. Tan, M.R. Islam and R.B.Roy , 'Probabilistic Load Flow-Based Optimal Placement and Sizing of Distributed Generators', Energies, MDPI, 14(23), 7857, November, 2021. [8] M. Rokonuzzaman, M.K. Mishu, N. Amin, M. Nadaraja, R. B. Roy , K.S. Rahman, A.M. Buhari, S.Binzaid, M. Shakeri and J. Pasupuleti, 'Self-Sustained Autonomous Wireless Sensor Network with Integrated Solar Photovoltaic System for Internet of Smart Home-Building (IoSHB) Applications', Micromachines, MDPI, 12(6), 653, June, 2021. [9] R. B. Roy , S. S. Alakoon and S.J. Arachcillage, 'Gird Impacts of Uncoordinated Fast Charging of Electric Ferries', Batteries, MDPI, 7(1), 13, February, 2021. [10] R. B. Roy and M. Rokonuzzaman, 'STATCOM Model for Integration of Wind Turbine to Grid' TELKOMNIKA Indonesian Journal of Electrical Engineering, ISSN: 2088-8708, Vol. 12, No. 9, pp. 6519 ~6525, September 2014. [11] F.A. Hossain, M. Rokonuzzaman, N. Amin, J. Zhang, M.K. Mishu, W.S. Tan, M.R. Islam and R.B.Roy , 'Probabilistic Load Flow-Based Optimal Placement and Sizing of Distributed Generators', Energies, 14(23), 7857, November, 2021. [12] M. R. Amin and R.B. Roy , 'Determination of Electric Vehicle Power Consumption for Assorted Pavement Gradient Using Matlab/Simulink', TELKOMNIKA Indonesian Journal

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of Electrical Engineering, vol. 12, no. 9, pp. 6602 ~ 6611, September, 2014.

- [13] M. R. Amin and **R.B. Roy**, 'Determination of Volume of Capacitor Bank for Static VAR Compensator', TELKOMNIKA Indonesian Journal of Electrical Engineering, vol. 12, no. 9, pp. 512-519, August, 2014.
- [14] **R.B. Roy** and M.R. Amin, 'A paper of Determination of Controlling Characteristics of Monopolar HVDC system', International Journal of Hybrid Information Technology, vol.7, no.3, pp. 97-112, 2014.
- [15] M. R. Amin, M.M. Hasan and **R.B. Roy**, 'Roadmap to Smart Grid Technology: A Review of Smart Information and Communication', International Journal of Control and Automation, vol. 12, no. 9, pp. 407-418, August, 2014.
- [16] M. R. Amin, **R.B. Roy** and M.M. Hasan, 'A comprehensive study of potential zone and low head hydro system in Bangladesh', International Journal of Innovative Technology and Creative Engineering, vol. 3, Issue 6, June, 2013.
- [17] M.J. Alam, H. Guoqing, M.R. Alam, M.G. Kibria and **R.B. Roy**, 'Improved ICT Grid model for development of Bangladesh', Applied Mechanics and Materials, vol. 411-414, pp 2200-2207, Trans Tech Publications, Switzerland, 2013.
- [18] M.R. Alam, **R.B. Roy**, M. M. Shiraj and H.M.M. Hasan, 'Trip Times Investigation of Ground Fault Circuit Interrupter', International Journal of Engineering and Computer Science, IJECS-IJENS, vol. 13, no. 6, 2013.
- [19] **R.B. Roy**, 'Design of Microcontroller Based Static VAR Compensator', International Journal of Scientific and Engineering Research, IJSER, vol.4, Issue 10, November 2012.
- [20] **R. B. Roy**, 'Design and Cost Analysis of Hybrid Power System for Off-Grid Rural Areas of Bangladesh', Canadian Journal on Electrical and Electronics Engineering, vol. 3, no.7, September. 2012.
- [21] **R. B. Roy**, 'Design and Performance Analysis of the Solar PV DC Water Pumping System', Canadian Journal on Electrical and Electronics Engineering, vol. 3, no.7, September, 2012.
- [22] **R. B. Roy** and M. R. Amin, 'Design and Construction of Single Phase Cycloconverter', International Journal of Recent Technology and Engineering (IJRTE), vol.1, issue 3, August, 2012.
- [23] M. R. Alam, **R. B. Roy**, M. J. Alam and D. J. Rahman, 'Single Phase Automatic Voltage Regulator Design for Synchronous Generator', International Journal of Electrical and Computer Sciences, vol. 11, no. 05, pp. 37-42, October, 2011.

Conference Publications

- [1] **R.B. Roy**, S. Alahakoon and P. J. V. Rensburg, 'Impacts of Uncoordinated Electric Ferry Charging on Distribution Network', 2024 International Conference on Sustainable Technology and Engineering (i-COSTE), December, 2024, IEEE.
- [2] **R.B. Roy**, S. Alahakoon, P. J. V. Rensburg, S. J. Arachchillage, 'Optimal Capacities of Electric Ferry Charging Stations by Hybrid Metaheuristic Algorithm', 34th Australian Universities Power Engineering Conference (AUPEC 2024), IEEE.
- [3] **R.B. Roy**, S. Alahakoon, P. J. V. Rensburg, S. J. Arachchillage, 'Analysis of Electric Ferry's Coordinated Charging Impacts on Distribution Network', 7th International Conference on Renewable Energy and Power Engineering (REPE, 2024), IEEE.
- [4] **R.B. Roy**, S. Alahakoon, P. J. V. Rensburg, E. Basher, R.M. Babu, 'Impacts on Distribution Network due to Coordinated Electric Ferry Charging', 3rd International Conference on Innovations in Science, Engineering and Technology (ICISSET 2024), IEEE.
- [5] **R.B. Roy**, S. Alahakoon, U. Mumtahina, A.A. Chowdhury, 'PSO Tuned Marine Microgrid Load Frequency Control', 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME), pp. 1-6, IEEE, 2022
- [6] **R.B. Roy**, S. Alakoon, 'Peak Shaving Approach of Distribution Network', International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON), pp.1-6, IEEE, 2022.
- [7] **R.B. Roy**, S. Alakoon, S. J. Arachchillage, 'Microgrid and Fleet to Grid Operation of a Hybrid Electric Ferry', Australasian Universities Power Engineering Conference, 2021, IEEE, Australia.
- [8] **R. B. Roy**, J. Cros, A. Nandi and T. Ahmed, 'Maximum Power Tracking by Neural Network', 8th International Conference on Reliability, Infocom Technologies and

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Optimization (Trends and Future Directions) (ICRITO), 4-5 June, 2020, IEEE, India.

- [9] **R. B. Roy**, E. Basher, S. M. B. Taslim and M. A. Hossain, 'Hybrid Power System Model for Off-Grid Rural Areas of Bangladesh', International Conference on Electrical, Communication, Electronics, Instrumentation and Computing (ICECEIC), 2019, IEEE.
- [4] **R. B. Roy**, J. Cros., E. Basher, D. S. Shawon and M. A. Islam, 'Precise Power Flow Analysis of MTDC', International Conference on Electrical, Communication, Electronics, Instrumentation and Computing (ICECEIC), 2019, IEEE, India.
- [5] **R. B. Roy**, J. Cros, E. Basher and S. M. B. Taslim, 'Fuzzy Logic Based Matrix Converter Controlled Induction Motor Drive', IEEE Region 10 Humanitarian Technology Conference (R10-HTC), 2017, IEEE, Bangladesh.
- [6] **R. B. Roy**, J. Cros, E. Basher and S. Akther, 'Power Compensation by DSTATCOM plus SCESS', 4th International Conference on Advances in Electrical Engineering (ICAEE), 2017, IEEE, Bangladesh.
- [7] **R. B. Roy**, M. Rokonzaman and M. H. E. Haider, 'Design and analysis of the power electronic transformer for power quality improvement', International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), 2015, IEEE, Bangladesh.
- [8] **R.B. Roy**, and M. R. Amin., 'Analysis of Controlling Topology of the Monopolar HVDC link between Bangladesh and India', 9th International Forum on Strategic Technology, IFOST, 2014, IEEE, Bangladesh.
- [9] **R. B. Roy**, E. Basher and J. M. Biswas, 'Design and Analysis of Isolated Integrated Charger for PHEV', 8th International conference on Electrical and Computer Engineering, ICECE 2014, IEEE, Bangladesh.
- [10] **R. B. Roy**, E. Basher, R. Yasmin and M. Rokonzaman, 'Fuzzy Logic Based MPPT Approach in a Grid Connected Photovoltaic System', 8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA 2014), IEEE, Bangladesh.
- [11] M. R. Amin and **R. B. Roy**, "Design and Simulation of Wireless Stationary Charging System for Hybrid Electric Vehicle using Inductive Power Pad in Parking Garage", 8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA 2014), IEEE, Bangladesh.
- [12] M. R. Amin and **R. B. Roy**., "Design of Microcontroller Based Thyristor Controlled Three-Phase Static Volt-Ampere Reactive Compensator", 3rd International conference on Informatics, Electronics and Vision'2014, IEEE, Bangladesh.

Best Conference Paper Award

R. B. Roy, E. Basher, R. Yasmin and M. Rokonzaman, 'Fuzzy Logic Based MPPT Approach in a Grid Connected Photovoltaic System', 8th International Conference on Software, Knowledge, Information Management and Applications (SKIMA 2014), IEEE, Bangladesh.

Research Grants	Ministry of Higher Education of Malaysia (MoHE) research grant, LRGS/1/2019/UKMUNITEN/6/2 Green Shore Power, GPC CQU MoU Project, Gladstone, Queensland, Australia, Document Number: 1890704
	Project Name: Green Shore Power This research was funded by the Gladstone Ports Corporation Limited (GPCL). GPC CQU MoU Project
	Project Name: Driving Resilience – Harnessing Green Mobile Energy Hubs for Electrical Distribution Network and Displaced Communities' Resilience Project Identifier: CQU.0001.2324D.RFI Funding Body: Queensland Reconstruction Authority (QRA) Funding Program: Queensland Resilience and Risk Reduction Fund (QRRRF)- 2023–24
Professional Memberships	IEB (Institute of Engineers, Bangladesh) IEEE (Institute of Electrical and Electronics Engineers)

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References	• Dr. Sanath Alahakoon, Senior Lecture and Discipline Leader, Department of Engineering and Technology, Central Queensland University, Queensland, Australia, Email: s.alahakoon@cqu.edu.au • Dipl.-Ing. Wulf Boie, Professor, Department of Energy and Environmental Management, University of Flensburg, Munketoft 3b, Flensburg, 24937, Germany. E-mail: boie@uni-flensburg.de • Dr. Olav Hohmeyer, Professor, Department for Sustainable Energy Transition, University of Flensburg, Munketoft 3b, Flensburg, 24937, Germany. E-mail: hohmeyer@uni-flensburg.de • Dr. Md. Rafiqul Alam, Professor, Department of Electrical and Electronic Engineering, Chittagong University of Engineering and Technology, Chittagong, Bangladesh. Contact No: 880-1621277369, Email: rabiulalam591@yahoo.com
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DR. RAJIB BARAN ROY, PhD

Tier-1 International Postdoctoral Research Portfolio

Professional Identity

ORCID: 0000-0002-1177-0778

Google Scholar: <https://scholar.google.com/citations?hl=en&user=oowfLzIAAAAJ>

Research Areas: Electrical Power Systems, Smart Grids, Renewable Energy Integration, Electric Transportation, Battery Energy Storage Systems, AI for Energy Systems

Executive Research Profile

Electrical Power Systems researcher with nearly two decades of academic experience and an established international publication record. Research expertise spans smart grids, electric ferry charging infrastructure, distribution network analysis, battery energy storage systems, renewable energy integration, microgrids, and AI-driven optimization. Extensive experience with OpenDSS, DIgSILENT PowerFactory, PSCAD, MATLAB/Simulink, and Python-based power system modelling.

Research Impact

Google Scholar Citations: 492

Google Scholar h-index: 13

Google Scholar i10-index: 16

Scopus Citations: 356

Scopus h-index: 7

Academic Book: 1

Journal Publications: 23

Conference Publications: 19+

International Research Grants: 3

ATS-Optimized Research Keywords

Power Systems, Smart Grids, Distribution Networks, Renewable Energy Integration, Electric Vehicles,

Electric Ferry Charging, Battery Energy Storage Systems, Energy Management Systems, Microgrids, Artificial Intelligence, Machine Learning, Metaheuristic Optimization, Power Quality, Grid Stability, Digital Twin, Vehicle-to-Grid, Renewable Hydrogen, Power System Resilience, Decarbonization, Net-Zero Energy Systems

Education

PhD in Electrical Engineering, Central Queensland University, Australia (2022–2026)
MSc in Sustainable Energy Systems and Management, University of Flensburg, Germany (2006–2008)

Academic Appointments

Assistant Professor, Primeasia University (2016–Present)
Senior Lecturer, University of Information Technology and Sciences (2009–2016)
Lecturer, University of Information Technology and Sciences (2004–2006)

Top 5 Most Influential Publications

1. Impact of Coordinated Electric Ferry Charging on Distribution Network Using Metaheuristic Optimization, *Energies* (2025)
2. Impact Analysis on Distribution Network due to Coordinated Electric Ferry Charging, *IET Energy Systems Integration* (2024)
3. Optimizing Dynamic Ferry Loads with Intelligent Power Management, *Energy Reports* (2023)
4. Comparative Performance Analysis of ANN Algorithms for MPPT Energy Harvesting in Solar PV Systems, *IEEE Access* (2021)
5. Grid Impacts of Uncoordinated Fast Charging of Electric Ferries, Batteries (2021)

Research Funding and Industry Collaboration

Green Shore Power Project – Gladstone Ports Corporation, Australia
Queensland Reconstruction Authority – Resilience Energy Hub Project
Ministry of Higher Education Malaysia LRGS Project

Collaborative research involving academia, government agencies, and industry stakeholders in energy transition and infrastructure resilience.

Future Research Agenda

Research Theme 1: AI-Driven Smart Distribution Networks
Research Theme 2: Electric Transportation Infrastructure Optimization
Research Theme 3: Battery Energy Storage and Grid Flexibility
Research Theme 4: Digital Twins for Power Systems
Research Theme 5: Coastal and Remote Community Energy Systems
Research Theme 6: Renewable Hydrogen and Sector Coupling
Research Theme 7: Net-Zero and Climate-Resilient Energy Infrastructure

Long-Term Vision:

Develop intelligent, resilient, and sustainable energy systems capable of supporting large-scale renewable integration and transportation electrification.

Two-Page Research Statement

My research focuses on the intersection of power systems, renewable energy integration, electric transportation, and artificial intelligence. The primary objective is to develop intelligent and resilient electricity networks capable of supporting future decarbonized energy systems.

My doctoral research investigates the impacts of fast-charging electric ferries on distribution networks and develops optimization strategies for charging coordination, energy storage integration, and network stability improvement. This work contributes to the emerging field of maritime electrification and sustainable transportation.

Future research will extend these contributions through AI-enabled decision support systems, digital twins, battery energy storage optimization, renewable hydrogen integration, and resilient coastal energy systems. The overarching goal is to enable reliable, economically viable, and environmentally sustainable electricity infrastructure for future communities.

The research methodology combines advanced power system modelling, optimization algorithms, machine learning, data analytics, and industry collaboration. Key software platforms include OpenDSS, DIgSILENT PowerFactory, PSCAD, MATLAB/Simulink, and Python.

Through international collaboration and interdisciplinary research, I aim to contribute to global energy transition initiatives and support the development of next-generation smart energy systems.

Master Postdoctoral Cover Letter

Dear Selection Committee,

I am applying for the Postdoctoral Research Fellow position in Electrical Engineering, Energy Systems, Smart Grids, or Renewable Energy. I am completing my PhD in Electrical Engineering at Central Queensland University, Australia, with research focused on electric ferry charging, distribution network impacts, battery energy storage systems, and optimization techniques.

My publication record includes 23 journal papers, 19+ conference papers, one academic book, and more than 492 Google Scholar citations. My research has appeared in leading journals including Energies, IEEE Access, Energy Reports, Batteries, Micromachines, and IET

Energy Systems Integration.

I have extensive experience using OpenDSS, DigSILENT PowerFactory, MATLAB/Simulink, PSCAD, and Python for power system analysis and optimization. In addition, I have participated in industry-funded and government-funded projects involving energy resilience, smart grids, and sustainable transportation.

I would welcome the opportunity to contribute to internationally recognized research programs addressing the challenges of energy transition, grid modernization, and transportation electrification.

Sincerely,

Dr. Rajib Baran Roy